inner mitochondrial membrane that transfer electrons from electron donors NADH and $FADH_2$ to acceptor such as molecular Oxygen. In the process, protons are pumped from the mitochondrial matrix to the inner membrane space, and eventually combine with O_2 and H^+ to form water (Figure 4.6).

HOTS

Why each NADH makes 3 ATPs and each FADH2 makes 2 ATPs?

As the electrons flow through the chain, much of their free energy is conserved in the form of ATP. The process by which energy from electron transport is used to make ATP is called as oxidative phosphorylation. Respiratory chain is an electron transport chain where a pair of electrons or hydrogen atoms containing electron from the substrate oxidized is coupled to reduction of oxygen to water.

The mitochondrial system is arranged into three complexes of electron carriers. They are

1. **Flavoproteins:** These proteins contain flavin, a coenzyme derived from riboflavin (Vit B_{12}). One important flavoprotein is flavin mono nucleotide.
2. **Ubiquinones (coenzyme Q):** These are small non protein carriers.
3. **Cytochromes:** These are proteins with iron containing group, capable of existing alternately as reduced (Fe^{2+}) and oxidized form (Fe^{3+}). Cytochromes involved in ETC include cyt (b),cyt c_1, cyt c, cyt a, cyt a_3.

The first step in electron transport chain is the transfer of high energy electrons from NADH to FMN. This transfer actually involves the passage of hydrogen atom with 2 e^- to FMN, which then picks up an additional H^+ from the surrounding aqueous medium. As

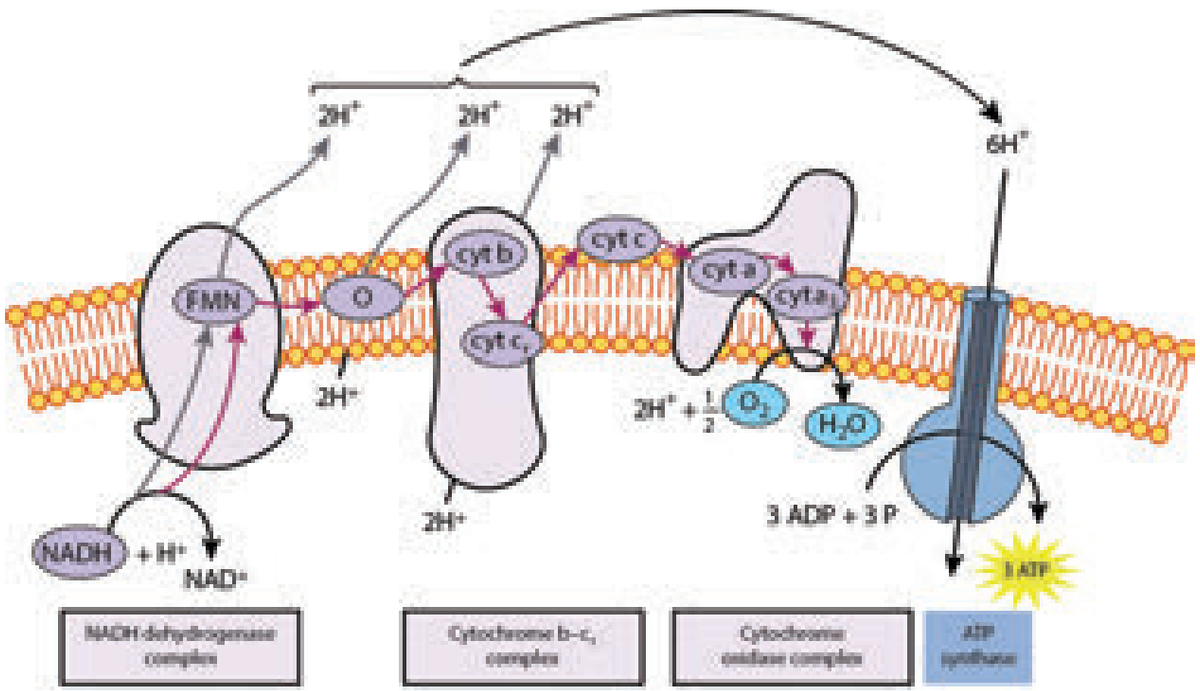


Figure 4.6: Electron Transport Chain and Chemiosmotic mechanism of ATP

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